EXTENSION 2

Stage 6

Calculus (Ext2), C1 Integration (Ext2)

Trig Integration

Teacher: Nardene Dodd

Exam Equivalent Time: 153 minutes *(based on allocation of 1.5 minutes per mark)*

Questions

1. Calculus, EXT2 C1 2020 HSC 6 MC

Which expression is equal to $\displaystyle\int \frac{1}{x^2+4x+10}\,dx$?

A. $\dfrac{1}{\sqrt{6}}\tan^{-1}\left(\dfrac{x+2}{\sqrt{6}}\right)+c$

B. $\tan^{-1}\left(\dfrac{x+2}{\sqrt{6}}\right)+c$

C. $\dfrac{1}{2\sqrt{6}}\ln\left|\dfrac{x+2-\sqrt{6}}{x+2+\sqrt{6}}\right|+c$

D. $\ln\left|\dfrac{x+2-\sqrt{6}}{x+2+\sqrt{6}}\right|+c$

2. Calculus, EXT2 C1 2013 HSC 1 MC

Which expression is equal to $\displaystyle\int \tan x\,dx$?

A. $\sec^2 x+c$

B. $-\ln(\cos x)+c$

C. $\dfrac{\tan^2 x}{2}+c$

D. $\ln(\sec x+\tan x)+c$

3. Calculus, EXT2 C1 2018 HSC 1 MC

Which expression is equal to $\displaystyle\int \frac{1}{\sqrt{1-4x^2}}\,dx$?

A. $\dfrac{1}{2}\sin^{-1}\dfrac{x}{2}+C$

B. $\dfrac{1}{2}\sin^{-1}2x+C$

C. $\sin^{-1}\dfrac{x}{2}+C$

D. $\sin^{-1}2x+C$

4. Calculus, EXT2 C1 2019 HSC 2 MC

Which of the following is a primitive of $\dfrac{\sin x}{\cos^3 x}$?

A. $\dfrac{1}{2}\sec^2 x$

B. $-\dfrac{1}{2}\sec^2 x$

C. $\dfrac{1}{4}\sec^4 x$

D. $-\dfrac{1}{4}\sec^4 x$

5. Calculus, EXT2 C1 2019 HSC 7 MC

Which of these integrals has the largest value?

A. $\displaystyle\int_0^{\frac{\pi}{4}} \tan x\,dx$

B. $\displaystyle\int_0^{\frac{\pi}{4}} \tan^2 x\,dx$

C. $\displaystyle\int_0^{\frac{\pi}{4}} 1-\tan x\,dx$

D. $\displaystyle\int_0^{\frac{\pi}{4}} 1-\tan^2 x\,dx$

6. Calculus, EXT2 C1 2022 HSC 4 MC

Of the following expressions, which one need NOT contain a term involving a logarithm in its anti-derivative?

A. $\dfrac{x+2}{x^2+4x+5}$

B. $\dfrac{x+2}{x^2-4x-5}$

C. $\dfrac{x-1}{x^3-x^2+x-1}$

D. $\dfrac{x+1}{x^3-x^2+x-1}$

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7. Calculus, EXT2 C1 2012 HSC 11c

By completing the square, find $\displaystyle\int \frac{dx}{x^2+4x+5}$. *(2 marks)*

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8. Calculus, EXT2 C1 2006 HSC 1b

By completing the square, find

$$\int \frac{dx}{x^2-6x+13}$$ *(2 marks)*

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9. Calculus, EXT2 C1 2007 HSC 1a

Find $\displaystyle\int \frac{1}{\sqrt{9-4x^2}}\,dx.$ *(2 marks)*

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10. Complex Numbers, EXT2 N1 2020 HSC 13d

i. Show that for any integer n, $e^{in\theta}+e^{-in\theta}=2\cos(n\theta)$. *(1 mark)*

ii. By expanding $\left(e^{i\theta}+e^{-i\theta}\right)^4$ show that

$$\cos^4\theta=\frac{1}{8}(\cos(4\theta)+4\cos(2\theta)+3).$$ *(3 marks)*

iii. Hence, or otherwise, find $\displaystyle\int_0^{\frac{\pi}{2}} \cos^4\theta\,d\theta$. *(2 marks)*

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11. Calculus, EXT2 C1 2021 SPEC2 2

Evaluate $\displaystyle\int_0^1 \frac{2x+1}{x^2+1}\,dx$. *(3 marks)*

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12. Calculus, EXT2 C1 2004 HSC 1c

By completing the square, find $\displaystyle\int \frac{dx}{\sqrt{5+4x-x^2}}$. *(2 marks)*

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13. Calculus, EXT2 C1 2019 HSC 11c

Find $\displaystyle\int \frac{dx}{x^2+10x+29}$ *(2 marks)*

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14. Calculus, EXT2 C1 2022 HSC 11b

Evaluate $\displaystyle\int \sin^3 2x\,\cos 2x\,dx$. *(2 marks)*

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15. Calculus, EXT2 C1 2022 HSC 11f

Using the substitution $t=\tan\dfrac{x}{2}$, find

$$\int \frac{dx}{1+\cos x-\sin x}$$ *(3 marks)*

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16. Calculus, EXT2 C1 2006 HSC 1e

Use the substitution $t=\tan\dfrac{\theta}{2}$ to show that

$$\int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{d\theta}{\sin\theta}=\frac{1}{2}\log 3.$$ *(3 marks)*

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17. Calculus, EXT2 C1 2007 HSC 1b

Find $\displaystyle\int \tan^2 x\sec^2 x\,dx.$ *(2 marks)*

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18. Calculus, EXT2 C1 2009 HSC 1c

Find $\displaystyle\int \frac{x^2}{1+4x^2}\,dx.$ *(3 marks)*

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19. Calculus, EXT2 C1 2010 HSC 1b

Evaluate $\displaystyle\int_0^{\frac{\pi}{4}} \tan x\,dx$. *(3 marks)*

20. Calculus, EXT2 C1 2010 HSC 1d

Using the substitution $t = \tan \dfrac{x}{2}$, or otherwise, evaluate $\displaystyle\int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x}$. *(4 marks)*

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21. Calculus, EXT2 C1 2011 HSC 1d

Find $\displaystyle\int \cos^3\theta \, d\theta$ *(3 marks)*

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22. Calculus, EXT2 C1 2011 HSC 1e

Evaluate $\displaystyle\int_{-1}^{1} \frac{1}{5-2t+t^2}\, dt.$ *(3 marks)*

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23. Calculus, EXT2 C1 2012 HSC 12a

Using the substitution $t = \tan \dfrac{\theta}{2}$, or otherwise, find $\displaystyle\int \frac{d\theta}{1-\cos\theta}$. *(3 marks)*

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24. Calculus, EXT2 C1 2013 HSC 12a

Using the substitution $t = \tan \dfrac{x}{2}$, or otherwise, evaluate

$\displaystyle\int_0^{\frac{\pi}{2}} \frac{1}{4+5\cos x}\, dx.$ *(4 marks)*

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25. Calculus, EXT2 C1 2017 HSC 11d

Using the substitution $t = \tan\dfrac{\theta}{2}$, or otherwise, evaluate

$\displaystyle\int_0^{\frac{2\pi}{3}} \frac{1}{1+\cos\theta}\, d\theta$. *(3 marks)*

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26. Calculus, EXT2 C1 2018 HSC 12c

Find $\displaystyle\int \frac{x^2+2x}{x^2+2x+5}\, dx$. *(3 marks)*

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27. Calculus, EXT2 C1 2018 HSC 14a

Using the substitution $t = \tan \dfrac{\theta}{2}$ evaluate $\displaystyle\int_0^{\frac{\pi}{2}} \frac{d\theta}{2-\cos\theta}$. *(3 marks)*

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28. Calculus, EXT2 C1 2021 HSC 12a

Find $\displaystyle\int \frac{2x+3}{x^2+2x+2} dx$. *(3 marks)*

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29. Calculus, EXT2 C1 SM-Bank 1

By completing the square and using the table of standard integrals, find

$\displaystyle\int \frac{dx}{4x^2-4x+10}$ *(2 marks)*

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30. Calculus, EXT2 C1 2005 HSC 1e

Let $t = \tan\left(\dfrac{\theta}{2}\right)$.

i. Show that $\dfrac{dt}{d\theta} = \dfrac{1}{2}\left(1+t^2\right)$ *(1 mark)*

ii. Show that $\sin\theta = \dfrac{2t}{1+t^2}$. *(2 marks)*

iii. Use the substitution $t = \tan\left(\dfrac{\theta}{2}\right)$ to find $\displaystyle\int \operatorname{cosec}\theta\, d\theta.$ *(2 marks)*

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31. Calculus, EXT2 C1 2015 HSC 11f

i. Show that

$\cot\theta + \operatorname{cosec}\theta = \cot\left(\dfrac{\theta}{2}\right).$ *(2 marks)*

ii. Hence, or otherwise, find

$\displaystyle\int (\cot\theta + \operatorname{cosec}\theta)\, d\theta.$ *(1 mark)*

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32. Calculus, EXT2 C1 2015 HSC 14a

i. Differentiate $\sin^{n-1}\theta\cos\theta$, expressing the result in terms of $\sin\theta$ only. *(2 marks)*

ii. Hence, or otherwise, deduce that

$\displaystyle\int_0^{\frac{\pi}{2}} \sin^n\theta\, d\theta = \frac{(n-1)}{n}\int_0^{\frac{\pi}{2}} \sin^{n-2}\theta\, d\theta$, for $n > 1.$ *(2 marks)*

iii. Find $\displaystyle\int_0^{\frac{\pi}{2}} \sin^4\theta\, d\theta.$ *(1 mark)*

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33. Calculus, EXT2 C1 2021 HSC 14a

Evaluate $\displaystyle\int_0^{\frac{\pi}{2}} \frac{1}{3+5\cos x}\, dx$. *(4 marks)*

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34. Calculus, EXT2 C1 2023 HSC 13a

Find $\displaystyle\int \frac{1-x}{\sqrt{5-4x-x^2}}\, dx$. *(3 marks)*

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35. Calculus, EXT2 C1 2014 HSC 13a

Using the substitution $t = \tan\dfrac{x}{2}$, or otherwise, evaluate

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{3\sin x - 4\cos x + 5}\,dx.$$ *(3 marks)*

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36. Calculus, EXT2 C1 2011 HSC 7b

Let $I = \displaystyle\int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)}\,dx.$

i. Use the substitution $u = 4 - x$ to show that

$$I = \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}u\right)}{u(4-u)}\,du.$$ *(2 marks)*

ii. Hence, find the value of I. *(3 marks)*

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37. Calculus, EXT2 C1 2013 HSC 14c

i. Given a positve integer n, show that

$$\sec^{2n}\theta = \sum_{k=0}^{n}\binom{n}{k}\tan^{2k}\theta.$$ *(1 mark)*

ii. Hence, by writing $\sec^8\theta$ as $\sec^6\theta\,\sec^2\theta$ find,

$$\int \sec^8\theta\,d\theta.$$ *(2 marks)*

Worked Solutions

1. Calculus, EXT2 C1 2020 HSC 6 MC

$$\begin{aligned}\int \frac{1}{x^3+4x+10}\,dx &= \int \frac{1}{(x+2)^2+\left(\sqrt{6}\right)^2}\,dx\\ &= \frac{1}{6}\tan^{-1}\left(\frac{x+2}{\sqrt{6}}\right)+c\end{aligned}$$

$\Rightarrow A$

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2. Calculus, EXT2 C1 2013 HSC 1 MC

$$\begin{aligned}\int \tan x\,dx &= \int \frac{\sin x}{\cos x}\,dx\\ &= -\ln(\cos x)+c\end{aligned}$$

$\Rightarrow B$

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3. Calculus, EXT2 C1 2018 HSC 1 MC

$$\frac{d}{dx}\left(\frac{1}{2}\sin^{-1}2x + C\right) = \frac{1}{\sqrt{1-4x^2}}$$

$\Rightarrow$ B

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4. Calculus, EXT2 C1 2019 HSC 2 MC

Solution 1

$$\begin{aligned}\int \frac{\sin x}{\cos^3 x} &= \int \sin x\cos^{-3}x\,dx\\ &= \frac{1}{2}\cos^{-2}x + C\\ &= \frac{1}{2}\sec^2 x + C\end{aligned}$$

Solution 2

$$\begin{aligned}\int \frac{\sin x}{\cos^3 x} &= \int \tan x\sec^2 x\,dx\\ &= \frac{1}{2}\tan^2 + C_1\\ &= \frac{1}{2}\left(\sec^2 x - 1\right) + C_1\\ &= \frac{1}{2}\sec^2 x + C_2\end{aligned}$$

$\Rightarrow A$

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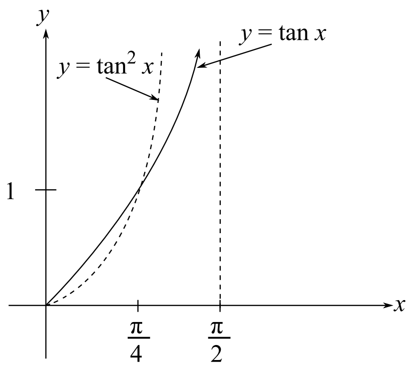
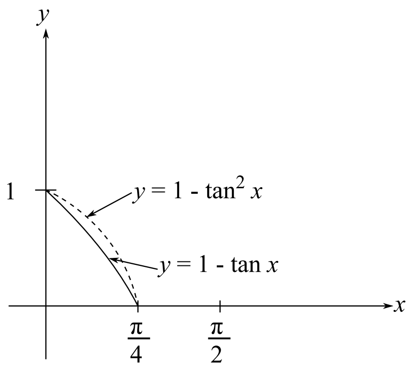
5. Calculus, EXT2 C1 2019 HSC 7 MC

Consider options A and B:



Consider options C and D:



$\therefore \int_0^{\frac{\pi}{4}} 1 - \tan^2 x \; dx$ is the largest

(largest area under the curve between $x = 0$ and $x = \frac{\pi}{4}$.)

$\Rightarrow D$

6. Calculus, EXT2 C1 2022 HSC 4 MC

Consider the denominator of C:

$$\begin{aligned} x^3 - x^2 + x - 1 &= x^2(x-1) + (x-1) \\ &= (x^2+1)(x-1) \end{aligned}$$

$$\frac{x-1}{x^3 - x^2 + x - 1} = \frac{x-1}{(x^2+1)(x-1)} = \frac{1}{x^2+1}$$

$$\int \frac{1}{x^2+1}\, dx = \tan^{-1}(x) + c$$

$\Rightarrow C$

7. Calculus, EXT2 C1 2012 HSC 11c

$$\begin{aligned} \int \frac{dx}{x^2+4x+5} &= \int \frac{dx}{x^2+4x+4+1} \\ &= \int \frac{dx}{(x+2)^2+1} \\ &= \tan^{-1}(x+2) + c \end{aligned}$$

8. Calculus, EXT2 C1 2006 HSC 1b

$$\begin{aligned} \int \frac{dx}{x^2 - 6x + 13} &= \int \frac{dx}{(x-3)^2 + 4} \\ &= \frac{1}{2} \tan^{-1}\left(\frac{x-3}{2}\right) + c \end{aligned}$$

9. Calculus, EXT2 C1 2007 HSC 1a

$$\begin{aligned} \int \frac{dx}{\sqrt{9 - 4x^2}} &= \frac{1}{2} \int \frac{dx}{\sqrt{\frac{9}{4} - x^2}} \\ &= \frac{1}{2} \sin^{-1} \frac{x}{\frac{3}{2}} + c \\ &= \frac{1}{2} \sin^{-1} \frac{2x}{3} + c \end{aligned}$$

10. Complex Numbers, EXT2 N1 2020 HSC 13d

i. $$\begin{aligned} e^{in\theta}+e^{-in\theta} &= \cos(n\theta)+i\sin(n\theta)+\cos(-n\theta)+i\sin(-n\theta)\\ &= \cos(n\theta)+i\sin(n\theta)+\cos(n\theta)-i\sin(n\theta)\\ &= 2\cos(n\theta)\end{aligned}$$

ii. $$\begin{aligned}\left(e^{i\theta}+e^{-i\theta}\right)^4 &= (2\cos\theta)^4\\ &= 16\cos^4\theta\end{aligned}$$

Expand $\left(e^{i\theta}+e^{-i\theta}\right)^4$:

$$\begin{aligned} & e^{i4\theta}+4e^{i3\theta}e^{-i\theta}+6e^{i2\theta}e^{-i2\theta}+4e^{i\theta}e^{-i3\theta}+e^{-i4\theta}\\ &= e^{i4\theta}+4e^{i2\theta}+6+4^{-i2\theta}+e^{-i4\theta}\\ &= e^{i4\theta}+e^{i4\theta}+4\left(e^{i2\theta}+e^{-i2\theta}\right)+6\\ &= 2\cos(4\theta)+8\cos(2\theta)+6\end{aligned}$$

$$\begin{aligned}\therefore\ 16\cos^4\theta &= 2\cos(4\theta)+8\cos(2\theta)+6\\ \cos^4\theta &= \frac{1}{8}\cos(4\theta)+\frac{1}{2}\cos(2\theta)+\frac{3}{8}\\ \cos^4\theta &= \frac{1}{8}(\cos(4\theta)+4\cos(2\theta)+3)\end{aligned}$$

iii. $$\begin{aligned}\int_0^{\frac{\pi}{2}}\cos^4\theta\ d\theta &= \frac{1}{8}\int_0^{\frac{\pi}{2}}\cos(4\theta)+4\cos(2\theta)+3\ d\theta\\ &= \frac{1}{8}\left[\frac{1}{4}\sin(4\theta)+2\sin(2\theta)+3\theta\right]_0^{\frac{\pi}{2}}\\ &= \frac{1}{8}\left[\left(\frac{1}{4}\sin(2\pi)+2\sin\pi+\frac{3\pi}{2}\right)-0\right]\\ &= \frac{1}{8}\left(\frac{3\pi}{2}\right)\\ &= \frac{3\pi}{16}\end{aligned}$$

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11. Calculus, EXT2 C1 2021 SPEC2 2

$$\begin{aligned}\int_0^1\frac{2x+1}{x^2+1}\ dx &= \int_0^1\frac{2x}{x^2+1}\ dx+\int_0^1\frac{1}{x^2+1}\ dx\\ &= \left[\log_e\left(x^2+1\right)\right]_0^1+\left[\tan^{-1}(x)\right]_0^1\\ &= \log_e 2-\log_e 1+\tan^{-1}(1)-\tan^{-1}(0)\\ &= \log_e 2+\frac{\pi}{4}\end{aligned}$$

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12. Calculus, EXT2 C1 2004 HSC 1c

$$\begin{aligned}\int\frac{dx}{\sqrt{5+4x-x^2}} &= \int\frac{dx}{\sqrt{9-(x^2-4x+4)}}\\ &= \int\frac{dx}{\sqrt{3^2-(x-2)^2}}\\ &= \sin^{-1}\left(\frac{x-2}{3}\right)+C\end{aligned}$$

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13. Calculus, EXT2 C1 2019 HSC 11c

$$\begin{aligned}\int\frac{dx}{x^2+10x+29} &= \int\frac{dx}{(x+5)^2+2^2}\\ &= \frac{1}{2}\tan^{-1}\left(\frac{x+5}{2}\right)+C\end{aligned}$$

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14. Calculus, EXT2 C1 2022 HSC 11b

$$\begin{aligned}\int\sin^3 2x\ \cos 2x\ dx &= \int\frac{1}{8}\times 4\times 2\cos 2x\times\sin^3 2x\ dx\\ &= \frac{1}{8}\sin^4 2x+c\end{aligned}$$

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15. Calculus, EXT2 C1 2022 HSC 11f

Let $t=\tan\dfrac{x}{2}$, $\cos x=\dfrac{1-t^2}{1+t^2}$, $\sin x=\dfrac{2t}{1+t^2}$

$dt=\dfrac{1}{2}\sec^2\dfrac{x}{2}\ dx\ \Rightarrow\ dx=\dfrac{2\ dt}{\sec^2\frac{x}{2}}=\dfrac{2}{1+t^2}\ dt$

$$\begin{aligned}\mathrm{I} &= \int\frac{dx}{1+\cos x-\sin x}\\ &= \int\frac{1}{1+\frac{1-t^2}{1+t^2}-\frac{2t}{1+t^2}}\cdot\frac{2}{1+t^2}\ dt\\ &= \int\frac{2}{1+t^2+1-t^2-2t}\ dt\\ &= \int\frac{1}{1-t}\ dt\\ &= -\ln|1-t|+C\\ &= -\ln\left|1-\tan\left(\frac{x}{2}\right)\right|+C\end{aligned}$$

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16. Calculus, EXT2 C1 2006 HSC 1e

Let $t = \tan\dfrac{\theta}{2},\quad \sin\theta = \dfrac{2t}{1+t^2},\quad d\theta = \dfrac{2}{1+t^2}\ dt$

When $\theta = \dfrac{\pi}{2},\quad t = 1$

When $\theta = \dfrac{2\pi}{3},\quad t = \sqrt{3}$

$$\begin{aligned}\therefore \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{d\theta}{\sin\theta} &= \int_1^{\sqrt{3}} \frac{1+t^2}{2t} \times \frac{2}{1+t^2}\ dt \\ &= \int_1^{\sqrt{3}} \frac{dt}{t} \\ &= [\ln t]_1^{\sqrt{3}} \\ &= \ln\sqrt{3} - \ln 1 \\ &= \ln 3^{\frac{1}{2}} - 0 \\ &= \frac{1}{2}\ln 3\end{aligned}$$

17. Calculus, EXT2 C1 2007 HSC 1b

$$\int \tan^2 x \sec^2 x\ dx = \frac{1}{3}\tan^3 x + c$$

18. Calculus, EXT2 C1 2009 HSC 1c

$$\begin{aligned}\frac{x^2}{1+4x^2} &= \frac{1}{4} \times \frac{4x^2}{1+4x^2} \\ &= \frac{1}{4} \times \frac{1+4x^2}{1+4x^2} - \frac{1}{4} \times \frac{1}{1+4x^2} \\ &= \frac{1}{4} - \frac{1}{4} \times \frac{1}{1+4x^2}\end{aligned}$$

$$\begin{aligned}\therefore \int \frac{x^2}{1+4x^2}\ dx &= \frac{1}{4}\int 1\ dx - \frac{1}{4}\int \frac{1}{1+4x^2}\ dx \\ &= \frac{x}{4} - \frac{1}{4} \times \frac{1}{2}\tan^{-1}2x + c \\ &= \frac{x}{4} - \frac{1}{8}\tan^{-1}2x + c\end{aligned}$$

19. Calculus, EXT2 C1 2010 HSC 1b

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \tan x\ dx &= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x}\ dx \\ &= [-\ln\cos x]_0^{\frac{\pi}{4}} \\ &= \left[-\ln\cos\frac{\pi}{4} - (-\ln\cos 0)\right] \\ &= -\ln\frac{1}{\sqrt{2}} + \ln 1 \\ &= \ln\sqrt{2} \\ &= \frac{1}{2}\ln 2\end{aligned}$$

20. Calculus, EXT2 C1 2010 HSC 1d

$t = \tan\dfrac{x}{2},\quad dx = \dfrac{2\ dt}{1+t^2},\quad \sin x = \dfrac{2t}{1+t^2}$

When $x = \dfrac{\pi}{2},\quad t = \tan\dfrac{\pi}{4} = 1$

When $x = 0,\quad t = \tan 0 = 0$

$$\begin{aligned}\therefore \int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x} &= \int_0^1 \frac{\frac{2\ dt}{1+t^2}}{1+\frac{2t}{1+t^2}} \\ &= \int_0^1 \frac{2}{1+t^2+2t}\ dt \\ &= \int_0^1 \frac{2}{(1+t)^2}\ dt \\ &= \left[\frac{-2}{1+t}\right]_0^1 \\ &= \frac{-2}{2} - \frac{-2}{1} \\ &= 1\end{aligned}$$

21. Calculus, EXT2 C1 2011 HSC 1d

$$\begin{aligned}\int \cos^3\theta\ d\theta &= \int \cos^2\theta\cos\theta\ d\theta \\ &= \int \left(1-\sin^2\theta\right)\cos\theta\ d\theta \\ &= \int \left(\cos\theta - \sin^2\theta\cos\theta\right) d\theta \\ &= \sin\theta - \frac{\sin^3\theta}{3} + c\end{aligned}$$

22. Calculus, EXT2 C1 2011 HSC 1e

$$\int_{-1}^{1} \frac{1}{(5-2t+t^2)} dt = \int_{-1}^{1} \frac{1}{(4+1-2t+t^2)dt}$$
$$= \int_{-1}^{1} \frac{1}{4+(t-1)^2} dt$$
$$= \frac{1}{2}\left[\tan^{-1}\left(\frac{t-1}{2}\right)\right]_{-1}^{1}$$
$$= \frac{1}{2}\left[\tan^{-1}0-\tan^{-1}(-1)\right]$$
$$= \frac{1}{2}\left[0-\left(-\frac{\pi}{4}\right)\right]$$
$$= \frac{\pi}{8}$$

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23. Calculus, EXT2 C1 2012 HSC 12a

$$t = \tan\frac{\theta}{2}$$
$$dt = \frac{1}{2}\sec^2\left(\frac{\theta}{2}\right) d\theta, \quad d\theta = \frac{2\ dt}{\sec^2\left(\frac{\theta}{2}\right)} = \frac{2}{1+t^2}\ dt$$
$$\cos\theta = \frac{1-t^2}{1+t^2}$$
$$\int \frac{d\theta}{1-\cos\theta} = \int \frac{1}{1-\left(\frac{1-t^2}{1+t^2}\right)} \times \frac{2}{1+t^2}\ dt$$
$$= \int \frac{2}{1+t^2-(1-t^2)}$$
$$= \int \frac{dt}{t^2}$$
$$= -\frac{1}{t} + c$$
$$= -\frac{1}{\tan\frac{\theta}{2}} + c$$
$$= -\cot\frac{\theta}{2} + c$$

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24. Calculus, EXT2 C1 2013 HSC 12a

$$t = \tan\frac{x}{2}$$
$$\Rightarrow \cos x = \frac{1-t^2}{1+t^2}\ , \quad dx = \frac{2\ dt}{1+t^2}$$
$$\text{When } x=0\ ,\ t=0\ ;\ \ x=\frac{\pi}{2}\ ,\ t=1$$
$$\int_0^{\frac{\pi}{2}} \frac{dx}{4+5\cos x} = \int_0^1 \frac{1}{4+\frac{5(1-t^2)}{1+t^2}} \times \frac{2\ dt}{1+t^2}$$
$$= \int_0^1 \frac{2\ dt}{4+4t^2+5-5t^2}$$
$$= \int_0^1 \frac{2\ dt}{9-t^2}$$
$$= 2\int_0^1 \frac{1}{(3-t)(3+t)}$$
$$= \frac{1}{3}\int_0^1 \left(\frac{1}{3-t}+\frac{1}{3+t}\right)\ dt$$
$$= \frac{1}{3}[-\ln(3-t)+\ln(3+t)]_0^1$$
$$= \frac{1}{3}[(-\ln 2+\ln 4)-(-\ln 3+\ln 3)]$$
$$= \frac{1}{3}\ln 2$$

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25. Calculus, EXT2 C1 2017 HSC 11d

Let $t = \tan\frac{\theta}{2},\ \cos\theta = \frac{1-t^2}{1+t^2},\ d\theta = \frac{2dt}{1+t^2}$

When $\theta = 0, t = \tan 0 = 0$

When $\theta = \frac{2\pi}{3}, t = \tan\frac{\pi}{3} = \sqrt{3}$

$$\int_0^{\frac{2\pi}{3}} \frac{1}{1+\cos\theta}\ d\theta = \int_0^{\sqrt{3}} \frac{\frac{2dt}{1+t^2}}{\frac{1+t^2}{1+t^2}+\frac{1-t^2}{1+t^2}}$$

$$= \int_0^{\sqrt{3}} \frac{\frac{2}{1+t^2}}{\frac{2}{1+t^2}}\ dt$$

$$= \int_0^{\sqrt{3}} 1\ dt$$

$$= [t]_0^{\sqrt{3}}$$

$$= \sqrt{3}-0$$

$$= \sqrt{3}$$

26. Calculus, EXT2 C1 2018 HSC 12c

$$\int \frac{x^2+2x}{x^2+2x+5}\ dx = \int \frac{(x^2+2x+5)-5}{x^2+2x+5}\ dx$$

$$= \int 1-\frac{5}{x^2+2x+5}\ dx$$

$$= \int 1-\frac{5}{2^2+(x+1)^2}\ dx$$

$$= x-\frac{5}{2}\tan^{-1}\left(\frac{x+1}{2}\right)+c$$

27. Calculus, EXT2 C1 2018 HSC 14a

$t = \tan\frac{\theta}{2},\ \cos\theta = \frac{1-t^2}{1+t^2},\ d\theta = \frac{2}{1+t^2}dt$

When $\theta = \frac{\pi}{2},\ t = 1$

When $\theta = 0,\ t = 0$

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{2-\cos\theta} = \int_0^1 \frac{1}{2-\frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2}\ dt$$

$$= \int_0^1 \frac{2}{2(1+t^2)-(1-t^2)}\ dt$$

$$= \int_0^1 \frac{2}{1+3t^2}\ dt$$

$$= \left[\frac{2}{\sqrt{3}}\tan^{-1}\left(\sqrt{3}t\right)\right]_0^1$$

$$= \frac{2}{\sqrt{3}}\tan^{-1}\sqrt{3}-0$$

$$= \frac{2\pi}{3\sqrt{3}}$$

$$= \frac{2\sqrt{3}\pi}{9}$$

28. Calculus, EXT2 C1 2021 HSC 12a

$$\int \frac{2x+3}{x^2+2x+2}dx = \int \frac{2x+2}{x^2+2x+2}dx + \int \frac{1}{(x-1)^2+1}dx$$

$$= \ln\left(x^2+2x+2\right)+\tan^{-1}(x+1)+c$$

29. Calculus, EXT2 C1 SM-Bank 1

$$\int \frac{dx}{4x^2-4x+10} = \int \frac{dx}{3^2+(2x-1)^2}$$

$$= \frac{1}{2}\int \frac{2}{3^2+(2x-1)^2}\ dx$$

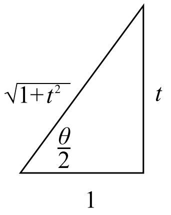
$$= \frac{1}{6}\tan^{-1}\left(\frac{2x-1}{3}\right)+C$$

30. Calculus, EXT2 C1 2005 HSC 1e

i. $t = \tan\dfrac{\theta}{2}$

$$\begin{aligned}\frac{dt}{d\theta} &= \frac{1}{2}\sec^2\frac{\theta}{2}\\ &= \frac{1}{2}\left(1+\tan^2\frac{\theta}{2}\right)\\ &= \frac{1}{2}(1+t^2)\end{aligned}$$

ii. Show $\sin\theta = \dfrac{2t}{1+t^2}$:



$$\begin{aligned}\sin\theta &= 2\,\sin\frac{\theta}{2}\cos\frac{\theta}{2}\\ &= 2\cdot\frac{t}{\sqrt{1+t^2}}\cdot\frac{1}{\sqrt{1+t^2}}\\ &= \frac{2t}{1+t^2}\end{aligned}$$

iii. $\displaystyle\int \operatorname{cosec}\,\theta\; d\theta$

$t = \tan\dfrac{\theta}{2}$

$$\frac{dt}{d\theta} = \frac{1}{2}\sec^2\frac{\theta}{2}\,,\; d\theta = \frac{2dt}{\sec^2\frac{\theta}{2}} = \frac{2}{1+t^2}\;dt$$

$$\begin{aligned}\int \operatorname{cosec}\,\theta\; d\theta &= \int\frac{1+t^2}{2t}\times\frac{2}{1+t^2}dt\\ &= \int\frac{1}{t}\;dt\\ &= \log_e|t|+c\\ &= \log_e\left|\tan\frac{\theta}{2}\right|+c\end{aligned}$$

---

31. Calculus, EXT2 C1 2015 HSC 11f

i.
$$\begin{aligned}\cot\theta+\operatorname{cosec}\,\theta &= \frac{\cos\theta}{\sin\theta}+\frac{1}{\sin\theta}\\ &= \frac{1+\cos\theta}{\sin\theta}\\ &= \frac{1+2\cos^2\left(\frac{\theta}{2}\right)-1}{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}\\ &= \frac{2\cos^2\left(\frac{\theta}{2}\right)}{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}\\ &= \frac{\cos\left(\frac{\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}\\ &= \cot\left(\frac{\theta}{2}\right)\end{aligned}$$

ii.
$$\begin{aligned}\int(\cot\theta+\operatorname{cosec}\,\theta)\;d\theta &= \int\cot\left(\frac{\theta}{2}\right)\;d\theta\\ &= \int\frac{\cos\left(\frac{\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}\;d\theta\\ &= 2\ln\left|\sin\,\frac{\theta}{2}\right|+c\end{aligned}$$

COMMENT: Part (ii) mean mark 51%.

---

32. Calculus, EXT2 C1 2015 HSC 14a

i. $\dfrac{d}{d\theta}\left(\sin^{n-1}\theta\cos\theta\right)$

$$\begin{aligned}
&= (n-1)\sin^{n-2}\theta\cos\theta\cos\theta + \sin^{n-1}\theta \times (-\sin\theta)\\
&= (n-1)\sin^{n-2}\theta\cos^2\theta - \sin^n\theta\\
&= (n-1)\sin^{n-2}\theta\left(1-\sin^2\theta\right) - \sin^n\theta\\
&= (n-1)\sin^{n-2}\theta - (n-1)\sin^n\theta - \sin^n\theta\\
&= (n-1)\sin^{n-2}\theta - n\sin^n\theta
\end{aligned}$$

ii. From part (i)

$$n\sin^n\theta = (n-1)\sin^{n-2}\theta - \frac{d}{d\theta}\left(\sin^{n-1}\theta\cos\theta\right)$$

$$\begin{aligned}
\therefore \int_0^{\frac{\pi}{2}} \sin^n\theta\ d\theta
&= \frac{1}{n}\int_0^{\frac{\pi}{2}} \left((n-1)\sin^{n-2}\theta - \frac{d}{d\theta}\left(\sin^{n-1}\theta\cos\theta\right)\right) d\theta\\
&= \frac{1}{n}\int_0^{\frac{\pi}{2}} (n-1)\sin^{n-2}\theta\ d\theta - \frac{1}{n}\int_0^{\frac{\pi}{2}} \frac{d}{d\theta}\left(\sin^{n-1}\theta\cos\theta\right)\ d\theta\\
&= \frac{1}{n}\int_0^{\frac{\pi}{2}} (n-1)\sin^{n-2}\theta\ d\theta - \frac{1}{n}\int_0^{\frac{\pi}{2}} \frac{d}{d\theta}\left(\sin^{n-1}\theta\cos\theta\right)\ d\theta\\
&= \frac{n-1}{n}\int_0^{\frac{\pi}{2}} \sin^{n-2}\theta\ d\theta - \frac{1}{n}\left[\sin^{n-1}\theta\cos\theta\right]_0^{\frac{\pi}{2}}\\
&= \frac{n-1}{n}\int_0^{\frac{\pi}{2}} \sin^{n-2}\theta\ d\theta - \frac{1}{n}(0-0)\\
&= \frac{n-1}{n}\int_0^{\frac{\pi}{2}} \sin^{n-2}\theta\ d\theta, \quad (n>1)
\end{aligned}$$

iii.
$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \sin^4\theta\ d\theta &= \frac{3}{4}\int_0^{\frac{\pi}{2}} \sin^2\theta\ d\theta\\
&= \frac{3}{4} \times \left[\frac{2-1}{2}\int_0^{\frac{\pi}{2}} d\theta\right]\\
&= \frac{3}{8} \times \left[\theta\right]_0^{\frac{\pi}{2}}\\
&= \frac{3\pi}{16}
\end{aligned}$$

---

33. Calculus, EXT2 C1 2021 HSC 14a

Let $t = \tan\dfrac{\theta}{2}$, $\cos\theta = \dfrac{1-t^2}{1+t^2}$

$$dt = \frac{1}{2}\sec^2\frac{\theta}{2}\ d\theta \ \Rightarrow\ d\theta = \frac{2\ dt}{\sec^2\frac{\theta}{2}} = \frac{2}{1+t^2}\ dt$$

Convert limits:

$\theta = \dfrac{\pi}{2} \ \rightarrow\ t = 1$

$\theta = 0 \ \rightarrow\ t = 0$

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \frac{1}{3+5\cos x}\ dx &= \int_0^1 \frac{1}{3+5\left(\frac{1-t^2}{1+t^2}\right)} \cdot \frac{2}{1+t^2}\ dt\\
&= \int_0^1 \frac{2}{3(1+t^2)+5(1-t^2)}\ dt\\
&= \int_0^1 \frac{1}{4-t^2}\ dt\\
&= \int_0^1 \frac{1}{(2+t)(2-t)}\ dt\\
&= \frac{1}{4}\int_0^1 \frac{1}{2+t} + \frac{1}{2-t}\ dt\\
&= \frac{1}{4}\left[\ln|2+t| - \ln|2-t|\right]_0^1\\
&= \frac{1}{4}\left[\ln 3 - \ln 1 - (\ln 2 - \ln 2)\right]\\
&= \frac{1}{4}\ln 3
\end{aligned}$$

34. Calculus, EXT2 C1 2023 HSC 13a

$$
\begin{aligned}
I &= \int \frac{1-x}{\sqrt{5-4x-x^2}}\,dx \\
&= \frac{1}{2}\int \frac{2-2x}{\sqrt{5-4x-x^2}}\,dx \\
&= \frac{1}{2}\int \frac{-4-2x}{\sqrt{5-4x-x^2}}\,dx + \frac{1}{2}\int \frac{6}{\sqrt{5-4x-x^2}}\,dx
\end{aligned}
$$

Let $u = 5-4x-x^2$

$$\frac{du}{dx} = -4-2x \;\Rightarrow\; du = -4-2x\,dx$$

$$
\begin{aligned}
I &= \frac{1}{2}\int u^{-\frac{1}{2}}\,du + 3\int \frac{1}{\sqrt{9-(x+2)^2}}\,dx \\
&= u^{\frac{1}{2}} + 3\sin^{-1}\left(\frac{x+2}{3}\right) + c \\
&= \sqrt{5-4x-x^2} + 3\sin^{-1}\left(\frac{x+2}{3}\right) + c
\end{aligned}
$$

---

35. Calculus, EXT2 C1 2014 HSC 13a

$$t = \tan\frac{x}{2}, \quad \sin x = \frac{2t}{1+t^2}$$

$$\cos x = \frac{1-t^2}{1+t^2}, \quad dx = \frac{2dt}{1+t^2}$$

$$\text{When} \quad x = \frac{\pi}{3}, \quad t = \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

$$\text{When} \quad x = \frac{\pi}{2}, \quad t = \tan\frac{\pi}{4} = 1.$$

$$
\begin{aligned}
&\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{3\sin x - 4\cos x + 5} \\
&= \int_{\frac{1}{\sqrt{3}}}^{1} \frac{1}{\frac{6t}{1+t^2} - \frac{4(1-t^2)}{1+t^2} + 5} \times \frac{2dt}{1+t^2} \\
&= \int_{\frac{1}{\sqrt{3}}}^{1} \frac{2}{6t-4+4t^2+5+5t^2}\,dt \\
&= \int_{\frac{1}{\sqrt{3}}}^{1} \frac{2}{9t^2+6t+1}\,dt \\
&= 2\int_{\frac{1}{\sqrt{3}}}^{1} \frac{dt}{(3t+1)^2} \\
&= -\frac{2}{3}\left[\frac{1}{3t+1}\right]_{\frac{1}{\sqrt{3}}}^{1} \\
&= -\frac{2}{3}\left(\frac{1}{4} - \frac{1}{\sqrt{3}+1}\right) \\
&= -\frac{2}{3}\left(\frac{1}{4} - \frac{\sqrt{3}-1}{2}\right) \\
&= -\frac{1}{6} + \frac{\sqrt{3}}{3} - \frac{1}{3} \\
&= \frac{2\sqrt{3}-3}{6}
\end{aligned}
$$

---

36. Calculus, EXT2 C1 2011 HSC 7b

i. Let $u = 4 - x, \quad du = -\,dx, \quad x = 4 - u$

When $x = 1, \quad u = 3$

When $x = 3, \quad u = 1$

$$\begin{aligned} I &= \int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)}\,dx \\ &= \int_3^1 \frac{\cos^2\left(\frac{\pi}{8}(4-u)\right)}{(4-u)u}(-\,du) \\ &= -\int_3^1 \frac{\cos^2\left(\frac{\pi}{2} - \frac{\pi}{8}u\right)}{(4-u)u}\,du \\ &= \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}u\right)}{u(4-u)}\,du \end{aligned}$$

ii. Since $\displaystyle\int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)}\,dx = \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}x\right)}{x(4-x)}\,dx$

We can add the integrals such that

$$\begin{aligned} 2I &= \int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)}dx + \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}x\right)}{x(4-x)}dx \\ &= \int_1^3 \frac{1}{x(4-x)}dx \end{aligned}$$

Using partial fractions:

$$\begin{aligned} \frac{1}{x(4-x)} &= \frac{A}{x} + \frac{B}{4-x} \\ 1 &= A(4-x) + Bx \end{aligned}$$

When $x = 0, \quad A = \frac{1}{4}$

When $x = 0, \quad B = \frac{1}{4}$

$$\begin{aligned} 2I &= \frac{1}{4}\int_1^3 \left(\frac{1}{x} + \frac{1}{4-x}\right)\,dx \\ &= \frac{1}{4}[\log_e x - \log_e(4-x)]_1^3 \\ &= \frac{1}{4}[\log_e 3 - \log_e 1 - (\log_e 1 - \log_e 3)] \\ &= \frac{1}{2}\log_e 3 \end{aligned}$$

$$\therefore I = \frac{1}{4}\log_e 3$$

♦ *Mean mark 36%.*

37. Calculus, EXT2 C1 2013 HSC 14c

i. $\sec^2\theta = 1 + \tan^2\theta$

$$\begin{aligned} \sec^{2n}\theta &= \left(1 + \tan^2\theta\right)^n \\ &= \binom{n}{0} + \binom{n}{1}\tan^2\theta + \ldots + \binom{n}{k}\tan^{2k}\theta + \\ &\qquad \ldots + \binom{n}{n}\tan^{2n}\theta \end{aligned}$$

$$\therefore \sec^{2n}\theta = \sum_{k=0}^{n}\binom{n}{k}\tan^{2k}\theta \quad \ldots \text{as required}$$

♦♦ *Mean mark 30%.*
MARKER'S COMMENT: Note that $\sec^{2n}\theta \backslash\backslash \neq 1 + \tan^{2n}\theta$. A very common error!

♦ *Mean mark 38%.*

ii.
$$\begin{aligned} \int \sec^8\theta\,d\theta &= \int \sec^6\theta\sec^2\theta\,d\theta \\ &= \int \left(1+\tan^2\theta\right)^3\sec^2\theta\,d\theta \\ &= \int \left[1 + \binom{3}{1}\tan^2\theta + \binom{3}{2}\tan^4\theta \right. \\ &\qquad \left. + \binom{3}{3}\tan^6\theta\right]\sec^2\theta\,d\theta \\ &= \int \left(1 + 3\tan^2\theta + 3\tan^4\theta + \tan^6\theta\right)\sec^2\theta\,d\theta \\ &= \tan\theta + \tan^3\theta + \frac{3}{5}\tan^5\theta + \frac{1}{7}\tan^7\theta + c \end{aligned}$$